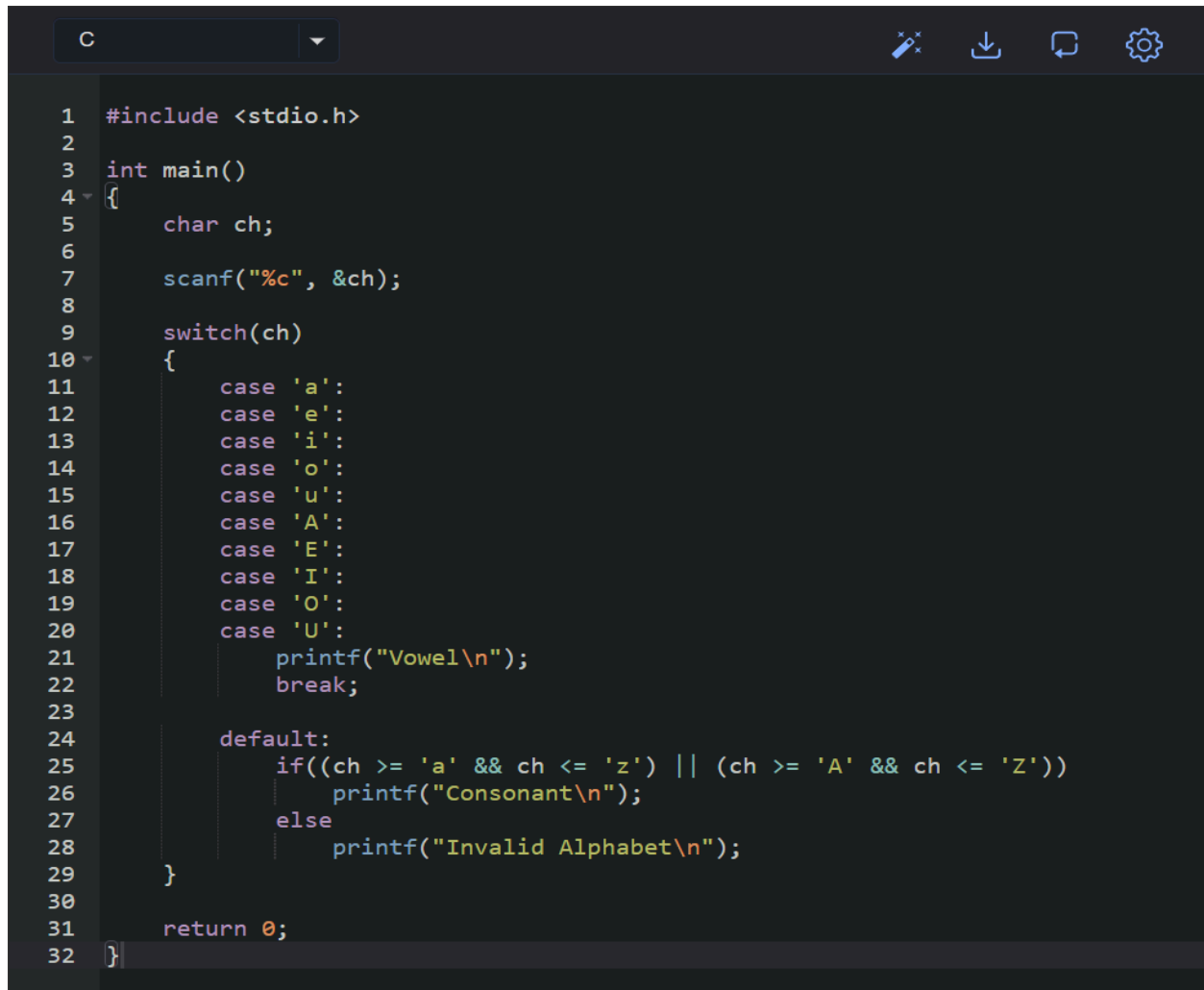


1. Write a program which takes a user input character value and prints if its vowel or consonant or any invalid alphabet using **switch** statement.

Syntax

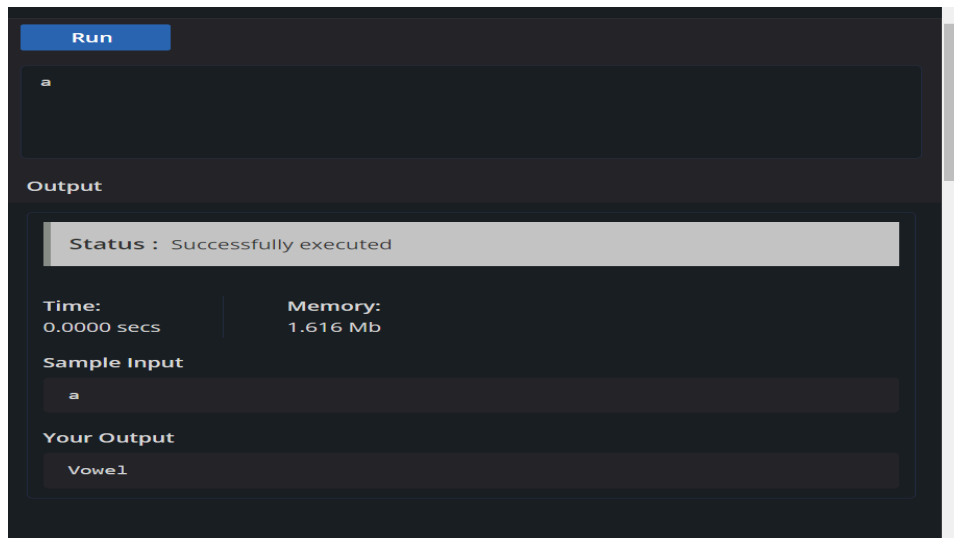
```
switch(variable) // variable can be of type int or char
{
    case value1 : statement 1
                break;
    case value2 : statement 2
                break;
    .
    .
    default:
}
}
```

Code:



```
C
1  #include <stdio.h>
2
3  int main()
4  {
5      char ch;
6
7      scanf("%c", &ch);
8
9      switch(ch)
10     {
11         case 'a':
12         case 'e':
13         case 'i':
14         case 'o':
15         case 'u':
16         case 'A':
17         case 'E':
18         case 'I':
19         case 'O':
20         case 'U':
21             printf("Vowel\n");
22             break;
23
24         default:
25             if((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z'))
26                 printf("Consonant\n");
27             else
28                 printf("Invalid Alphabet\n");
29     }
30
31     return 0;
32 }
```

Output:



---

2. Write a program using the **switch** statement that will Read 2 numbers and an operator (+, -, \*, /) . Based on the operator, perform the corresponding operation and print the result. (Calculator)

Eg. Enter two numbers

4      5

If user input is +

Output = 9

Code: Please scroll down,

```

1  #include <stdio.h>
2
3  int main()
4  {
5      float num1, num2, result;
6      char op;
7
8      scanf("%f %f", &num1, &num2);
9      scanf(" %c", &op);
10
11     switch(op)
12     {
13         case '+':
14             result = num1 + num2;
15             printf("Output = %.2f\n", result);
16             break;
17
18         case '-':
19             result = num1 - num2;
20             printf("Output = %.2f\n", result);
21             break;
22
23         case '*':
24             result = num1 * num2;
25             printf("Output = %.2f\n", result);
26             break;
27
28         case '/':
29             result = num1 / num2;
30             printf("Output = %.2f\n", result);
31             break;
32
33         default:
34             printf("Invalid Operator\n");
35     }
36
37     return 0;
38 }

```

Output:

4 5  
+

Output

Status : Successfully executed

Time: 0.0000 secs

Memory: 1.748 Mb

Sample Input

4 5  
+

Your Output

Output = 9.00

3. Write a program using the **switch** statement to read a single digit no (0..9) and print the corresponding word (Zero, One .. Nine etc,)

Code:

```
C
1  #include <stdio.h>
2
3  int main()
4  {
5      int num;
6
7      scanf("%d", &num);
8
9      switch(num)
10     {
11         case 0:
12             printf("Zero\n");
13             break;
14         case 1:
15             printf("One\n");
16             break;
17         case 2:
18             printf("Two\n");
19             break;
20         case 3:
21             printf("Three\n");
22             break;
23         case 4:
24             printf("Four\n");
25             break;
26         case 5:
27             printf("Five\n");
28             break;
29         case 6:
30             printf("Six\n");
31             break;
32         case 7:
33             printf("Seven\n");
34             break;
35         case 8:
36             printf("Eight\n");
37             break;
38         case 9:
39             printf("Nine\n");
40             break;
41         default:
42             printf("Invalid Digit\n");
43     }
44
45     return 0;
46 }
```

Output:

Output

Status : Successfully executed

|             |          |
|-------------|----------|
| Time:       | Memory:  |
| 0.0000 secs | 1.792 Mb |

Sample Input

5

Your Output

Five

4. Write a program using the **switch** statement to print the number of days in a month using switch case.

Code:

```
C
1  #include <stdio.h>
2
3  int main()
4  {
5      int month;
6
7      scanf("%d", &month);
8
9      switch(month)
10     {
11         case 1:
12             printf("31 days\n");
13             break;
14
15         case 2:
16             printf("28 or 29 days\n");
17             break;
18
19         case 3:
20             printf("31 days\n");
21             break;
22
23         case 4:
24             printf("30 days\n");
25             break;
26
27         case 5:
28             printf("31 days\n");
29             break;
30
31         case 6:
32             printf("30 days\n");
33             break;
34
35         case 7:
```

```

27     case 5:
28         printf("31 days\n");
29         break;
30
31     case 6:
32         printf("30 days\n");
33         break;
34
35     case 7:
36         printf("31 days\n");
37         break;
38
39     case 8:
40         printf("31 days\n");
41         break;
42
43     case 9:
44         printf("30 days\n");
45         break;
46
47     case 10:
48         printf("31 days\n");
49         break;
50
51     case 11:
52         printf("30 days\n");
53         break;
54
55     case 12:
56         printf("31 days\n");
57         break;
58
59     default:
60         printf("Invalid Month\n");
61 }
62
63 return 0;
64 }

```

Output:

Run

4

Output

Status : Successfully executed

Time: 0.0000 secs

Memory: 1.704 Mb

Sample Input

4

Your Output

30 days

5. Write a program to print the colors of rainbow based on the user input character.

case 'v':

case 'V': printf("\n Violet"); break;

..

case 'r':

case 'R': printf("\n Red"); break;

default: printf("\n Color is not in rainbow");

Code:

```
C
1  #include <stdio.h>
2
3  int main()
4  {
5      char ch;
6
7      scanf("%c", &ch);
8
9      switch(ch)
10     {
11         case 'v':
12         case 'V':
13             printf("Violet\n");
14             break;
15
16         case 'i':
17         case 'I':
18             printf("Indigo\n");
19             break;
20
21         case 'b':
22         case 'B':
23             printf("Blue\n");
24             break;
25
26         case 'g':
27         case 'G':
28             printf("Green\n");
29             break;
30
31         case 'y':
32         case 'Y':
33             printf("Yellow\n");
34             break;
35     }
```

```

13         printf("Violet\n");
14         break;
15
16     case 'i':
17     case 'I':
18         printf("Indigo\n");
19         break;
20
21     case 'b':
22     case 'B':
23         printf("Blue\n");
24         break;
25
26     case 'g':
27     case 'G':
28         printf("Green\n");
29         break;
30
31     case 'y':
32     case 'Y':
33         printf("Yellow\n");
34         break;
35
36     case 'o':
37     case 'O':
38         printf("Orange\n");
39         break;
40
41     case 'r':
42     case 'R':
43         printf("Red\n");
44         break;
45
46     default:
47         printf("Color is not in rainbow\n");
48     }
49     return 0;
50 }
51

```

Output:

Run

v

Output

Status : Successfully executed

Time: 0.0000 secs

Memory: 1.684 Mb

Sample Input

v

Your Output

Violet



